Olivier Côté ACAS

PhD Candidate in Actuarial Science
Université Laval

EMAIL olivier.cote.12@ulaval.ca
WEB olicoside.github.io
LINKEDIN in/olivier-cote-act
LANGUAGES Français (native), English (C1), Svenska (A1)

Education

PhD in Actuarial Science (GPA 4.17/4.33) 2023–Exp. 2027
Université Laval
Fairness in insurance pricing, supervised by Prof. Marie-Pier Côté (Université Laval) and Prof. Arthur Charpentier (UQÀM).

Master's degree in Actuarial Science, research-based (GPA 4.24/4.33), Université Laval 2021–2023
Bachelor's degree in Actuarial Science (GPA 4.17/4.33), Université Laval 2018–2021

Research visits 2023–2026
- Department of Mathematics, Stockholm University (2026): seminar and discussions with Prof. Mathias Millberg Lindholm.
- Department of Statistics, University of Toronto (2025): seminar and discussions with Prof. Christopher Blier-Wong.
- Bayes Business School, London (2023): discussions with Prof. Andreas Tsanakas.

Research papers

Côté, O., Côté, M.-P., and Charpentier, A. (2027). A statistical perspective on fairness in insurance pricing. *Annual Review of Statistics and Its Application*. Invited review, accepted June 2026. In press

Charpentier, A., **Côté, O.**, and Côté, M.-P. (2026b). Governing fairness in insurance pricing: The trilemma of actuarial adequacy, solidarity, and causal legitimacy. *Journal of Business Ethics*. doi:10.1007/s10551-026-06479-2

Côté, M.-P., **Côté, O.**, and Charpentier, A. (2026). Recoverability of market-wide fair insurance premiums under selection bias. *Insurance: Mathematics and Economics*, 129:103268. doi:10.1016/j.insmatheco.2026.103268

Côté, O., Côté, M.-P., and Charpentier, A. (2026). A scalable toolbox for exposing indirect discrimination in insurance rates. *CAS Research Paper Series on Bias and Insurance*. Research paper, casact.org
- Winner of the American Academy of Actuaries Award for Research 2026
- Selected by the National Association of Insurance Commissioners (NAIC) for its webinar series for U.S. regulators

Charpentier, A., Fernandes Machado, A., **Côté, O.**, and Hu, F. (2026a). Federated measurement of demographic disparities from quantile sketches. *arXiv:2602.18870*. Preprint

Côté, O., Côté, M.-P., and Charpentier, A. (2025). A fair price to pay: Exploiting causal graphs for fairness in insurance. *Journal of Risk and Insurance*, 92(1):33–75. doi:10.1111/jori.12503
- Finalist for the American Academy of Actuaries Award for Research 2025

Work in progress

Côté, O., Charpentier, A., Côté, M.-P., and Il Idrissi, M. A definition of confounding bias for arbitrary statistical functionals. Working paper, in preparation.

Côté, O., Côté, M.-P., Charpentier, A., and Il Idrissi, M. Profiling actuarial discrimination via causal decomposition. Working paper, presented at the 29th International Congress on Insurance: Mathematics and Economics, Seoul.

Peer review

European Actuarial Journal 2026–present
Annals of Actuarial Science 2026–present
Journal of Risk and Insurance 2024–present

Diversity, equity and inclusion education

Get Comfortable Being Uncomfortable: Dialogue About Favoritism and Fairness, CANSSI · · · 2025
Let's talk about race and collective responsibility, CANSSI · · · 2024
Acting as a sentinel for the prevention of suicide, Centre de prévention du suicide du Québec · · · 2023
Addressing Conflicts Related to Bias, Privilege, and Identity in the STEM Fields, CANSSI · · · 2022

Work experience

Course lecturer, two mandatory undergraduate courses (3 credits each) · · · 2022–2023
Université Laval

- Linear Models in Actuarial Science (fall 2022, 88 students)
- Statistical Learning in Actuarial Science (winter 2023, 54 students)

Awards: Faculty of Science and Engineering Teaching Star, one per course (fall 2022 and winter 2023).

Online course on fair insurance pricing, project lead and script writer · · · 2025–2026
Casualty Actuarial Society (English and French versions, in post-production)

Commissioned by the CAS. Coordinated four parties: a research assistant, the CAS task force on biases in insurance pricing, the CAS educational development team and a production company. Two hours of two-minute videos, activities and quizzes, co-presented with M.-P. Côté and A. Charpentier.

American football coach · · · 2019–2025
Falcons of Cégep de Lévis

American football coach for different teams for more than ten years (2012–2025), including seven years in collegiate football at the Cégep de Lévis (2019–2025):

- Tight ends and Fullbacks coach (2019–2025), responsible for the development of 10 athletes
- Supervisor for a bachelor's degree intern in sports intervention (2023)
- Volunteer mathematics tutor for student-athletes

Awards: RSEQ Coaching Staff of the Year 2023 and 2024, Sportsmanship Award 2022 (D1) and 2023 (D2).

Teaching assistant · · · 2019–2024
Université Laval

Statistical Learning in Actuarial Science (5 times), Probabilistic Analysis of Actuarial Risks (3 times), Linear Models in Actuarial Science (once), internship evaluation (once).

Actuarial intern · · · 2019–2021
Desjardins General Insurance Group

Four internships: property ratemaking (2019), business analysis (2020 and 2021) and research (2021).

Supervision

Co-supervisor of a PhD student, Université Laval · · · 2025–present
With Prof. Marie-Pier Côté, Mitacs Accelerate partnership

Participated in negotiating the student's Mitacs funding with the industrial partner and in designing the thesis project.

Co-supervisor of a master's student, Université Laval · · · 2023
With Prof. Marie-Pier Côté

One-year project on fairness in insurance. The student now works as an actuary.

Professional designations

Associate of the Casualty Actuarial Society (ACAS), Casualty Actuarial Society · · · 2025
Professionalism Workshop, Canadian Institute of Actuaries · · · 2025
MAS-I, MAS-II, Exams 5 and 6C, Online Courses 1 and 2, CAS · · · 2020–2025
Exams P, FM and IFM, Society of Actuaries · · · 2019–2020

Scholarships and awards

Award for Research · · · 2026
American Academy of Actuaries
Winner (honorarium US$7 500, ≈ 75 000 SEK) for *A scalable toolbox for exposing indirect discrimination in insurance rates*. Annual international competition for papers by early-career scholars.

Travel grants · · · 2026
NSERC, American Academy of Actuaries, Big Data Research Center
Three competitive grants for conference travel: NSERC General Graduate Studies Fund ($1 300 CAD, ≈ 9 100 SEK), American Academy of Actuaries (US$1 500, ≈ 15 000 SEK) and Big Data Research Center ($2 000 CAD, ≈ 14 000 SEK).

Canada Graduate Research Scholarship – Doctoral · · · 2025–2028
Natural Sciences and Engineering Research Council of Canada
National competition, $120 000 CAD (≈ 841 000 SEK) over three years, held from May 2025.

Hickman Scholars Program · · · 2024–2027
Society of Actuaries
International competition, US$20 000 (≈ 199 000 SEK) per year for three years (US$60 000 in total, ≈ 598 000 SEK). Every annual renewal is competitive.

Doctoral Training Scholarships · · · 2024–2025
Fonds de recherche du Québec – Nature et technologies
Provincial competition, $100 000 CAD (≈ 701 000 SEK) offered over four years, held from May 2024 to May 2025, when the NSERC doctoral scholarship began.

Mitacs Accelerate · · · 2024–2027
Mitacs
Research partnership with a large Canadian P&C insurer, $84 000 CAD (≈ 589 000 SEK) over three years (2024–2027), in which I am the lead student researcher. Selected after a scientific peer review and an internal Mitacs review.

Doctorate attraction grant Florent Toureille · · · 2023–2026
Chaire d'actuariat de l'Université Laval
Institutional competition based on leadership and excellence, $30 000 CAD (≈ 210 000 SEK) over three years.

Leadership and Engagement Scholarship · · · 2023–2026
Université Laval
Institutional competition in the sports sector, $30 000 CAD (≈ 210 000 SEK) over three years.

Charles Brindamour Graduate Scholarship for Excellence · · · 2023
Université Laval, funded by Charles Brindamour (CEO of Intact Financial Corporation)
Institutional competition based on excellence, one award a year, $10 000 CAD (≈ 70 000 SEK).

François-Laviolette Scholarship · · · 2022
Université Laval
Institutional competition for responsible artificial intelligence, $7 500 CAD (≈ 53 000 SEK).

Grant of attraction to the Master's · · · 2021
Chaire d'actuariat de l'Université Laval
Institutional competition recognizing research potential, $15 000 CAD (≈ 105 000 SEK).

CAS Trust Scholarship · · · 2021
Casualty Actuarial Society
International competition, US$2 500 (≈ 25 000 SEK).

Alexander Graham Bell Canada Graduate Scholarship – Master's · · · 2021–2022
Natural Sciences and Engineering Research Council of Canada
National competition, $17 500 CAD (≈ 123 000 SEK).

Lieutenant Governor's Youth Medal · · · 2018
Cégep de Lévis
Awarded for outstanding involvement and academic results, on nomination by the teaching staff.

Academic communications

Operationalizing Fairness in Actuarial Pricing: From Principle to Practice · · · 2025–2026
Casualty Actuarial Society and University of Toronto
Sole presenter at a 90-minute CAS webinar (2026, 950 registrants). Presented at the CAS Annual Meeting in Austin (2025, over 150 participants) and at the CAS Canada Connect Day in Toronto (2025, over 100). Seminar at the Department of Statistics, University of Toronto (2025). *Audience: practitioners and academics.*

A Scalable Toolbox for Exposing Indirect Discrimination in Insurance Rates · · · 2025–2026
International conferences and invited talks
Presented at the first ASTIN Bulletin Conference in Zurich (2026) and at PARTY in Liverpool (2025, 60 doctoral students from 29 countries). Presented in French with Arthur Charpentier at the Chaire ACTIONS and Milliman R&D in Paris (2025, 80 participants). *Audience: academics and practitioners.*

Profiling Actuarial Discrimination via Causal Decomposition · · · 2026
Conferences and seminars
Talk at the 29th International Congress on Insurance: Mathematics and Economics in Seoul and poster at the Actuarial and Financial Mathematics Conference in Brussels. Seminar at STATQAM, UQÀM, also covering confounding bias, and research talk at the Big Data Research Center, Université Laval. *Audience: academics.*

The Trilemma of Fairness in P&C Pricing · · · 2026
American Academy of Actuaries and AAIARD
Invited session of the American Academy of Actuaries at the 61st Actuarial Research Conference. Presented in French to the Association des actuaires en IARD du Québec as *Le trilemme de l'équité en tarification IARD*. *Audience: decision-makers and practitioners.*

Bias in Assessing Financial Risk: Insights from the 2025 Research Award · · · 2026
American Academy of Actuaries
Invited panelist as a 2025 finalist. Webinar to 445 actuaries and practitioners. *Audience: practitioners.*

A Causal Perspective on Fairness in Insurance Pricing · · · 2026
Department of Mathematics, Stockholm University
Seminar during research stay. *Audience: academics.*

Two invited presentations at the 60th Actuarial Research Conference · · · 2025
American Academy of Actuaries panel & Casualty Actuarial Society session, Toronto
Guest panelist and presenter with Marie-Pier Côté and Arthur Charpentier. *Audience: decision-makers.*

A Fair Price to Pay: Exploiting Causal Graphs for Fairness in Insurance · · · 2024
Conference and workshop
Talk at the 27th International Congress on Insurance: Mathematics and Economics in Chicago. Presented to over 60 academics, practitioners and regulators at the Workshop on fairness and discrimination in insurance. *Audience: academics.*

Perspective causale sur la discrimination indirecte en assurance générale · · · 2024
Autorité des marchés financiers
Presentation to decision-makers at Québec's financial authority, followed by interviews with ten regulators on their needs. *Audience: decision-makers.*

Perspective causale sur les discriminations dans les modèles prédictifs en assurance · · · 2024
Institut intelligence et données
Guest presenter to over 100 people from academia. *Audience: academics.*

Équité et discrimination en assurance générale · · · 2023
Intact Financial Corporation
Presentation to the CEO, CLO and CDO. *Audience: decision-makers.*

Causal Inference and Fairness in Insurance Pricing · · · 2023
Insurance Data Science Conference, London
Conference presentation. *Audience: academics.*

Public outreach

L'avenir à travers une boule de cristal mathématique : Le travail d'un actuaire 2023–2026
Cégep de Lévis
Guest presenter for CEGEP students. *Audience: students.*

Pour une intelligence artificielle équitable en assurance 2024
Université Laval News
I collaborated for a scientific communication article. *Audience: general public.*

Perspective causale sur la discrimination indirecte en assurance générale 2024
Institut Intelligence et données
4th research competition on data intelligence. ≈ 50 graduate students. *Audience: general public.*

Perspective causale sur la discrimination indirecte en assurance générale 2024
Institut de valorisation des données
I presented my research during IVADO Digital Future Day 2024. Over 250 people. *Audience: general public.*

L'IA pour calculer le prix de votre assurance auto: poétique ou pas éthique? 2023
Big Data Research Center
Presentation during the event *lightning talks 180*. *Audience: general public.*
Awards: Public's choice award

Artificial intelligence to prevent unfair discrimination in auto insurance 2023
IVADO
I authored an 800-word scientific communication article. *Audience: general public.*

Event organization

Workshop on fairness and discrimination in insurance 2026
Université Laval, 6–7 October 2026, 90 participants
Member of the organizing committee, host of the second day, panel moderator and presenter.

Quantact Student Day 2025
Province-wide event, 3 October 2025
Organizer: recruited a five-member committee, coordinated seven presenters and hosted sixty students.

Committees

Institut intelligence et données, Université Laval, board member 2025–present
Graduate Program Committee in Actuarial Science, Université Laval, voting member 2022–present
Big Data Research Center, Université Laval, student voting member 2022–2025
Actuarial Science Student Investment Fund, Université Laval, voting member 2021–2024

Research affiliations

Institut Intelligence et Données, Big Data Research Center, Université Laval 2021–present
Quantact research laboratory, Centre de recherches mathématiques, Montréal 2021–present

Other

Roundtable on raising awareness about youth bullying 2025
Mammouth hebdo - Téléquébec

Actuarial School representative at the open house 2022–2023
Faculty of Science and Engineering
About 50 visits from students and their parents in fall 2022, about 30 in winter 2023.